

Dr. Rajesh Kumar Yadav

Research publication portfolio

Affiliation

Department of Physics, S. K. M. University, Dumka, India

Research areas

SUSY quantum mechanics; non-Hermitian quantum theory; mathematical physics

Scholar metrics

361 citations; h-index 13; i10-index 13

Coverage

22 works listed on the public profile, 2013–2026

Profile

[Google Scholar record](#)

Year-wise publication matrix

Rows are shaded by complete year blocks. Position is the researcher’s place in the abbreviated author list (position/total). Click an arXiv number to open the PDF.

Year	Author pos.	Authors (abbrev.)	Paper title	Received / preprint*	Published	Journal / publication	arXiv PDF
2026	1/3	R. K. Yadav; R. Kumar; A. Khare	Rational extension of quantum anisotropic oscillator potentials with linear and/or quadratic perturbations	2025-09-05	2025-12-11 (3.2 mo)	International Journal of Modern Physics A 41(01), 2650003	2504.08236
2025	2/3	R. Kumar; R. K. Yadav; A. Khare	Rational extension of anisotropic harmonic oscillator potentials in higher dimensions	2024-11-05*	2025-08-01 (~8.8 mo*)	Annals of Physics 479, 170045	2411.02955
2024	2/4	S. Banerjee; R. K. Yadav; A. Khare; B. P. Mandal	A class of exactly solvable real and complex PT symmetric reflectionless potentials	2023-09-03	2024-03-01 (5.9 mo)	Journal of Mathematical Physics 65(3)	2308.05917
2024	2/2	S. Banerjee; R. K. Yadav	One continuous parameter family of Dirac Lorentz scalar potentials associated with exceptional orthogonal polynomials	2023-09-22	2024-02-06 (4.5 mo)	International Journal of Modern Physics A 38(35–36), 2350184	2309.12965
2024	2/3	R. Kumar; R. K. Yadav; A. Khare	Rationally extended harmonic oscillator potential, isospectral family and the uncertainty relations	2023-04-22*	2024-04 (~1.0 yr*)	Annals of Physics 463, 169623	2304.11314
2023	2/5	S. Banerjee; R. K. Yadav; A. Khare; N. Kumari; B. P. Mandal	Solutions of (1+1)-dimensional Dirac equation associated with exceptional orthogonal polynomials and the parametric symmetry	2023-01-25	2023-07-11 (5.5 mo)	International Journal of Modern Physics A 38, 2350069-1	2211.02557
2022	2/2	S. Banerjee; R. K. Yadav	Scattering state solutions of a rationally extended PT symmetric complex potential using potential algebra approach	2022-03-25	2022-08-19 (4.8 mo)	Mathematical Statistician and Engineering Applications 71(4), 8340–8350	–
2022	1/5	R. K. Yadav; S. Banerjee; N. Kumari; A. Khare; B. P. Mandal	One parameter family of rationally extended isospectral potentials	2020-04-28*	2022-01-01 (~1.7 yr*)	Annals of Physics 436, 168679	2004.13478
2019	1/4	R. K. Yadav; A. Khare; N. Kumari; B. P. Mandal	Rationally extended many-body truncated Calogero–Sutherland model	2018-07-13*	2019-01 (~6 mo*)	Annals of Physics 400, 189	1807.05163

Continued on next page

Year	Author pos.	Authors (abbrev.)	Paper title	Received / preprint*	Published	Journal / publication	arXiv PDF
2018	2/4	N. Kumari; R. K. Yadav; A. Khare; B. P. Mandal	A class of exactly solvable rationally extended non-central potentials in two and three dimensions	2017-07-10*	2018-06-07 (~10.9 mo*)	Journal of Mathematical Physics 59, 062103	1707.03289
2017	2/4	N. Kumari; R. K. Yadav; A. Khare; B. P. Mandal	A class of exactly solvable rationally extended Calogero–Wolfe type three-body problems	2017-03-29*	2017-10 (~6 mo*)	Annals of Physics 385, 57–69	1703.10161
2017	2/4	S. Bera; R. K. Yadav; B. Chakrabarti; B. P. Mandal	Role of conditional shape invariance symmetry property to obtain eigen-spectrum of the generalized polynomial potential with a Coulomb term	2017-06-01*	–	arXiv preprint	1706.00168
2017	6/6	A. Ramos; B. Bagchi; A. Khare; N. Kumari; B. P. Mandal; R. K. Yadav	A short note on “Group theoretic approach to rationally extended shape invariant potentials”	2017-05-11*	2017-07 (~2 mo*)	Annals of Physics 382, 143	1705.04592
2017	1/4	R. K. Yadav; N. Kumari; A. Khare; B. P. Mandal	Rationally extended shape invariant potentials in arbitrary D -dimensions associated with exceptional polynomials	2014-12-16*	2017-12-30 (~3.0 yr*)	Acta Polytechnica 57(6), 477–484	1412.5445
2016	2/5	N. Kumari; R. K. Yadav; A. Khare; B. Bagchi; B. P. Mandal	Scattering amplitudes for the rationally extended PT symmetric complex potentials	2016-04-08*	2016-10 (~6 mo*)	Annals of Physics 373, 163–177	1604.02452
2016	1/5	R. K. Yadav; A. Khare; B. Bagchi; N. Kumari; B. P. Mandal	Parametric symmetries in exactly solvable real and PT symmetric complex potentials	2015-10-05*	2016-06-30 (~8.8 mo*)	Journal of Mathematical Physics 57, 062106	1510.01666
2015	1/4	R. K. Yadav; N. Kumari; A. Khare; B. P. Mandal	Group theoretic approach to rationally extended shape invariant potentials	2015-02-26*	2015-08 (~5 mo*)	Annals of Physics 359, 46–54	1502.07455
2015	1/1	R. K. Yadav	Symmetries and their consequences in quantum theories associated with exceptional orthogonal polynomials	–	2015	Doctoral thesis, Banaras Hindu University	–
2013	1/3	R. K. Yadav; A. Khare; B. P. Mandal	The scattering amplitude for rationally extended shape invariant Eckart potentials	2013-09-26*	2015-01 (~1.3 yr*)	Physics Letters A 379, 67–70	1309.6755
2013	1/3	R. K. Yadav; A. Khare; B. P. Mandal	The scattering amplitude for one parameter family of shape invariant potentials related to X_m Jacobi polynomials	2013-03-15*	2013-06-25 (~3.4 mo*)	Physics Letters B 723(4), 433–435	1303.3669
2013	3/3	B. P. Mandal; B. K. Mourya; R. K. Yadav	PT phase transition in higher-dimensional quantum systems	2013-01-11*	2013-06 (~5 mo*)	Physics Letters A 377(14), 1043–1046	1301.2387
2013	1/3	R. K. Yadav; A. Khare; B. P. Mandal	The scattering amplitude for a newly found exactly solvable potential	–	2013-04	Annals of Physics 331, 313–316	–

Reading notes. Alternating pale-blue and white bands apply to complete profile-year groups. An unstarred first date is the publisher’s received/submitted date, and the bracketed duration is publisher turnaround to online publication. A starred date means the journal’s received date was unavailable, so the first arXiv date is used; its starred bracket is only a preprint-to-publication interval, not publisher turnaround. Month-only publication dates produce approximate month durations. A dash means that the corresponding date or arXiv record was not identified. Profile years and author order follow the displayed Google Scholar record; therefore a profile year can differ from the later journal publication year.

Data snapshot. Public Google Scholar profile and linked arXiv records, checked 21 July 2026. Scholar metrics are dynamic and may change after this snapshot.